



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Computer Vision

Chapter 0: General information

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About me

- Dr. Nguyen Thi Oanh
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- Office:
 - 706 - B1 (working office) / 1002-B1
- Teaching:
 - Computer vision, image processing
 - Databases, database labs
 - Intro do DS, Intro to ICT
- Research:
 - Semantic segmentation (on medical images)
 - Domain adaptation for semantic segmentation
 - Action recognition (with multi-view, multi-modality)
 - Image representation and retrieval



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General information

- Course name:

COMPUTER VISION

- Code: IT5409
- Credit: 3(3-1-0-6)
 - Lecturer: 45 hours
 - Capstone project: 15 hours
 - Experiments: 0 hours

Evaluation

- Mid-term (0.4)
 - Capstone project evaluation
 - Program
 - Report
 - Presentation
 - Bonus
- Final term: (0.6)
 - Multi-choice questions
 - Short/long answer questions

Rules

- In-class attendance
- Telephone:
 - turn-off or in vibration mode
- Come in/ go out if necessary
 - No need for asking permission
 - Without noise

Course Content

- **Chapter 1.** Introduction
- **Chapter 2.** Image formation, acquisition and digitization
- **Chapter 3.** Image Processing
- **Chapter 4.** Feature detection and matching
- **Chapter 5.** Segmentation
- **Chapter 6.** Motion object detection and tracking
- **Chapter 7.** Object recognition and deep learning

How to learn?

- Class attendance
- Reading additional articles / books
- Practice your-self (OpenCV, ...)
- QA
- Communication channel:
 - MS Teams (see class code on qltdt.hust.edu.vn)

Reference books

- [1]. Richard Szeliski (2011). Computer Vision: Algorithms and Applications. Springer.
<http://szeliski.org/Book/>
- [2]. David A. Forsyth, Jean Ponce (2011). Computer Vision: A modern Approach. Pearson
- [3]. Ranjay Krishna, Ed and Compiler "Computer Vision: Foundations and Application", Stanford University, First printing, December 2017.



